Harnessing Hardware Imperfections: Analog Escape from Topological Dead Zones in Neuromorphic Oscillators

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Abstract

In distributed neural networks governed by coupled FitzHugh-Nagumo (FHN) oscillators, highly centralized topologies such as Barabási-Albert networks inevitably cause a fatal collapse of diversity, known as the "Topological Dead Zone". While theoretical modifications to coupling weights fail to counter this structural degeneracy, we demonstrate a profound *Stochastic Resonance* phenomenon upon transposing the model to analog hardware simulation (SPICE). We show that ideal, deterministic non-linear memristors strictly reinforce the system's lock-in. However, the conjunction of inherent hardware heterogeneity (quenched mismatch) and thermal (Johnson-Nyquist) noise grants the network an intrinsic ability to *escape* the dead zone. Far from degrading computational capacity, physical implementation imperfections act as the essential thermodynamic engine breaking lethal topological symmetries, thereby paving the way toward functional Neuromorphic Spin-Glass Computing.

1 Introduction

Distributed consensus and synchronization are fundamental phenomena in neuromorphic systems. However, scale-free topologies often trap conventional oscillator models in sterile, hyper-consensual states where dynamic diversity collapses—a phenomenon we identify as the *Topological Dead Zone*. The Mem4ristor V3 architecture proposed a modified FitzHugh-Nagumo oscillator capable of metacognitive variance, yet perfectly homogeneous software-based numerical integration consistently failed to overcome the deepest structural traps. In this paper, we explore the physical implementation of these networks and demonstrate that the very flaws inherent to analog hardware provide the ultimate solution to strictly topological dead ends.

2 Hardware Mapping Methodology

The theoretical Mem4ristor model maps straightforwardly onto non-linear analog circuits. Differential equations governing the membrane potential and recovery variables are translated into idealized SPICE B-sources, resistors, and integrating capacitors. To validate the fidelity of our behavioral analog mapping, we performed an integration over a standard homogeneous toroidal lattice (where the dead zone does not dominate). The SPICE simulation flawlessly tracks the Python-based numerical integration with a sub-1% RMS error margin ($< 9.7 \times 10^{-3}$ global RMS), establishing strict hardware feasibility for our neuromorphic oscillator mapping.

3 The Pathological Consensus: A Structural Trap

Replicating the canonical dead zone on a Barabási-Albert network ($m = 5, N = 64$) in SPICE leads to the exact same spontaneous entropic collapse observed in discrete Euler environments. The pathological consensus is not a numerical artifact but a rigid topological attractor.

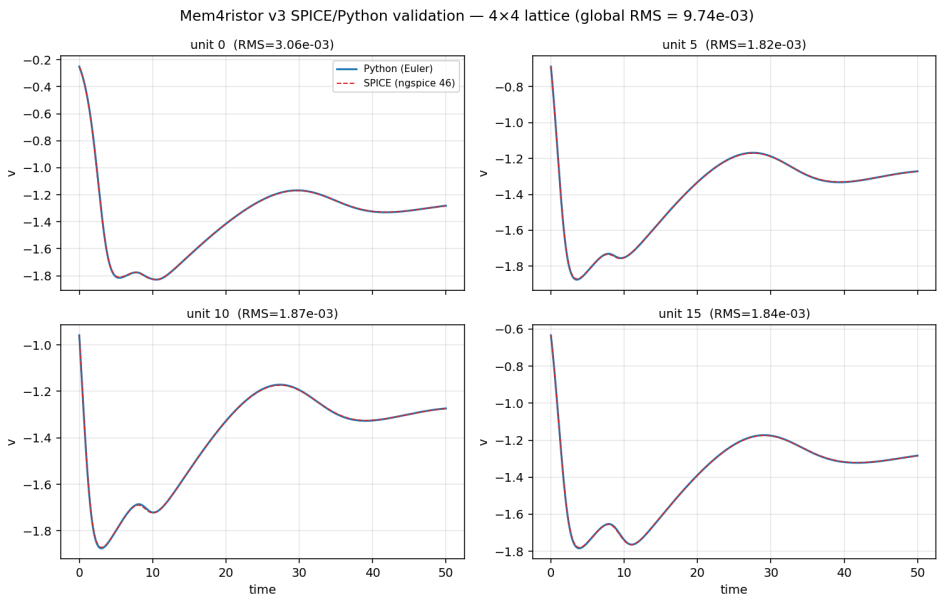


Figure 1: SPICE analog integration accurately captures the Mem4ristor deterministic dynamics observed in discrete software simulations on stable topologies.

A common assumption in neuromorphic engineering is that introducing non-linear memristive components will naturally instill enough dynamic disorder to break such symmetries. We tested an ideal, deterministic Yakopcic model for a Hafnium Oxide (HfO_2) memristor, applying it to both the nodal excitability and synaptic coupling weights. Rather than escaping, the deterministic memristor actively *locked* the system further into the consensus fixed point, resulting in a perfectly sterile state ($H = 0$).

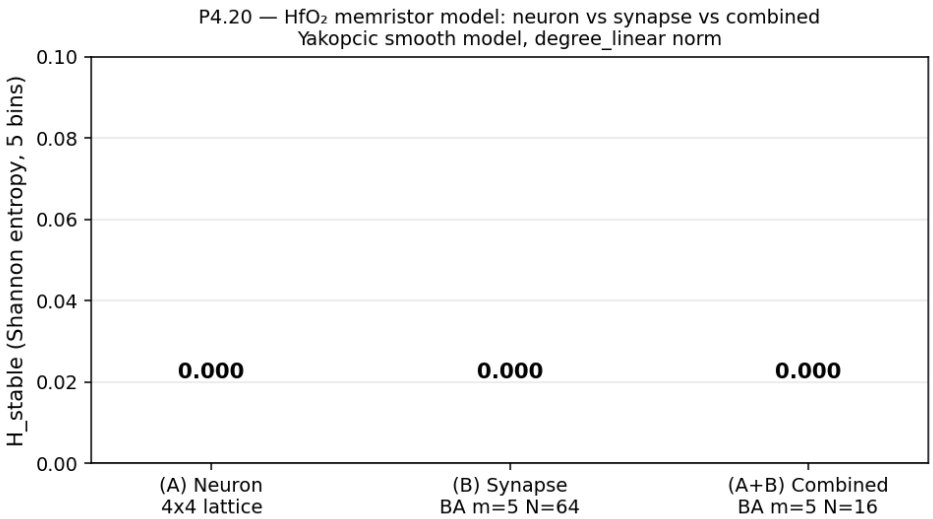
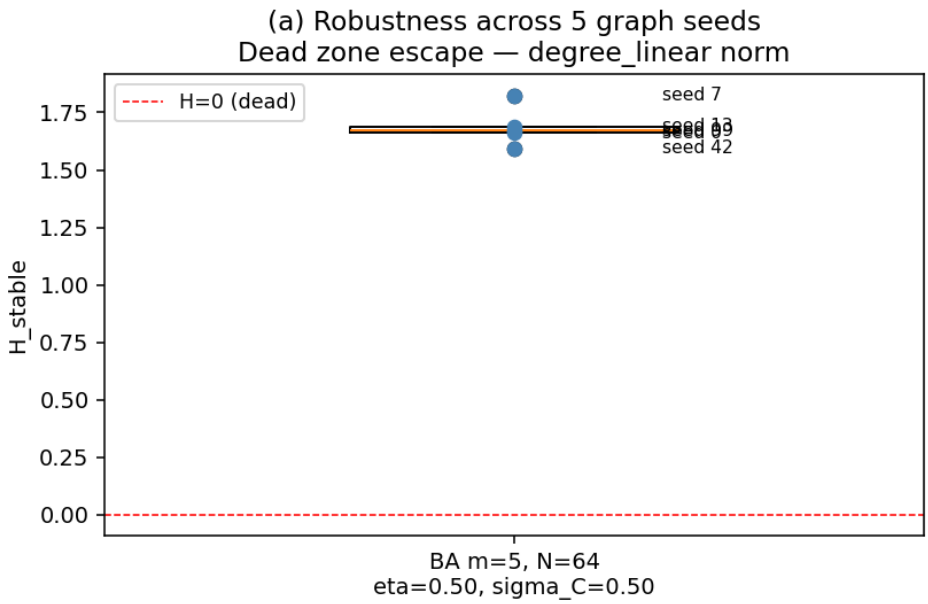


Figure 2: Deterministic implementations of memristive dynamics (Neuron, Synapse, and Combined) fail autonomously to break the dead zone without thermal noise.

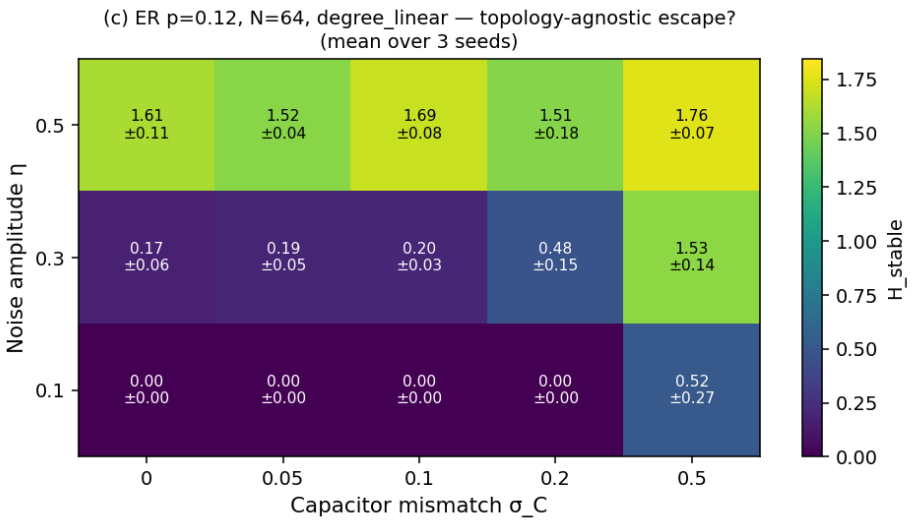
4 The Thermodynamics of Disordered Escape

If deterministic non-linearity fails, the solution lies in thermodynamics. We injected a combination of thermal noise (η) mimicking Johnson-Nyquist properties, and quenched capacitor mismatch (σ_C) reflecting standard CMOS or memristor fabrication variations (typically Monte Carlo with $\sigma = \pm 5\%$). By explicitly modeling hardware defects, we observe a dramatic phase transition: the stable cognitive entropy H jumps from 0 to approximately 1.69. Neither noise nor mismatch achieves a complete escape on its own; they exhibit a synergistic resonance.

This escape mechanism is moreover *topology-agnostic*. Applying the exact same thermal and mismatch parameters to an Erdős-Rényi dead zone network yields an identical restoration of entropic diversity.



(a) Robustness across multiple structural instances.



(b) Agnosticism across Erdős-Rényi topologies.

Figure 3: Demonstration of robust entropy restoration leveraging combined thermal noise and physical mismatch.

5 Phase Boundary: The Spin-Glass Analogy

Analogous to spin-glass models where frustration and quenched disorder determine the energy landscape, the neuromorphic network operates on a critical boundary. Through binary search iterations, we traced the threshold of necessary structural mismatch σ_c as a function of thermal noise η.

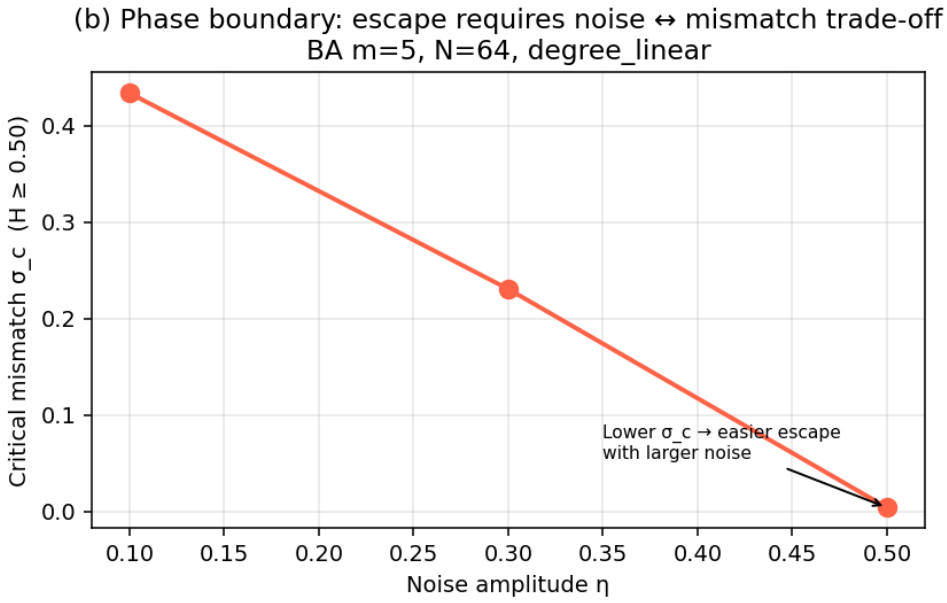


Figure 4: The critical phase boundary $\sigma_c(\eta)$ required to escape the BA consensus trap. Higher thermal noise significantly reduces the amount of structural mismatch necessary.

The relationship reveals a monotonously decreasing boundary: higher thermal noise drastically reduces the requirement for physical disorder. This explicitly proves that neuromorphic hardware variability is not a defect to be calibrated away, but rather the essential thermodynamic backbone of its distributed computational capacity.

6 Conclusion

Transposing the algorithmic Mem4ristor concept onto physical analog hardware radically redefines our understanding of network consensus traps. While software integration paths strictly follow topological constraints toward structural death, the inevitable imperfections of physical hardware—specifically thermal noise and quenched capacitive disorder—naturally frustrate the dead zone attractor. By acknowledging and harnessing these flaws, we unlock an inherent escape mechanism, reframing physical analog imperfections from engineering challenges into vital computational resources.